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KLE Technological University, Hubballi.



A Minor Project Report
on

A Drift-Aware Scheduling in 5G RAN

submitted in partial fulfillment of the requirement for the degree of

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Computer Science and Engineering
(AI)

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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that Mini Project titled **A Drift-Aware Scheduling in 5G RAN** is a bonafide work carried out by the student team comprising of Akshat Dodwad – 01FE23BCI045, Naveen Huggi – 01FE23BCI056, Abhishek M – 01FE23BCI116, Riya A Magadum – 01FE23BCI096 for partial fulfillment of completion of fifth semester B.E. in Computer Science and Engineering(AI) during the academic year 2025-26.

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ABSTRACT

The dynamic nature of 5G networks, characterized by diverse services and varying channel conditions, often renders static scheduling algorithms ineffective due to "concept drift." This leads to suboptimal Quality of Service (QoS), including reduced throughput and unfair resource distribution. To address this, we propose a "Drift-Aware Scheduling" framework within the Open Radio Access Network (O-RAN) architecture. Leveraging the Near-Real-Time RIC, we introduce a Machine Learning (ML) xApp that uses an unsupervised Isolation Forest algorithm to monitor Key Performance Indicators (KPIs) like throughput and PRB usage. Unlike supervised methods, our approach detects anomalies in real-time from sliding windows of telemetry data. Upon detecting drift (sudden or gradual), the system autonomously tunes MAC-layer scheduling parameters via the E2 interface. Implemented on a NIST-based O-RAN testbed with OpenAirInterface and FlexRIC, experimental results show that our drift-aware scheduler achieves a 15% improvement in average user throughput and maintains a Jain's Fairness Index of 0.99 under congestion. This demonstrates the viability of autonomous, data-driven optimization for 5G networks.

Keywords: 5G, O-RAN, Scheduling, Drift Detection, Isolation Forest, Machine Learning, FlexRIC.

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Chapter 1

INTRODUCTION

1.1 Motivation

The motivating factor behind this project is the need to bridge the gap between fixed network rules and the dynamic nature of real-world 5G environments. While 5G networks are designed to support high-performance applications, their management systems often rely on static schedules that cannot easily adapt to change.

Traditional Radio Access Networks (RAN) typically use rigid scheduling algorithms. These algorithms work well when the network traffic is stable and predictable. However, in a real-world scenario, user behavior is constantly changing. For instance, a sudden surge in data demand at a specific location can cause congestion that a static scheduler is not programmed to handle. This lack of flexibility leads to lower data speeds and increased delay (latency), preventing the network from delivering its promised quality of service.

To address these limitations, modern networks are beginning to use Machine Learning (ML) models. These models learn from past data to make better decisions. However, they face a significant challenge known as *concept drift*. Concept drift occurs when the patterns of network traffic change over time, making the data used to train the model outdated. If an ML model continues to make decisions based on old patterns, its performance will degrade. Detect and correcting this drift is essential for maintaining a reliable network.

There is a clear need for a scheduling system that is capable of autonomous adaptation. Instead of relying on manual updates or fixed rules, the system should be able to monitor itself and detect when its performance is dropping. By integrating an unsupervised drift detection mechanism into the Open RAN (O-RAN) architecture, this project aims to create a scheduler that can sense changes in the environment and adjust its strategy in real-time. This approach ensures that the network remains efficient and fair for all users, regardless of how the traffic conditions evolve.

1.2 Literature Review

Al-Zubaidy et al. (2021) proposed a beam division multiple access (BDMA) based scheduling algorithm to improve capacity in congested 5G scenarios. The approach dynamically splits beams to serve multiple users simultaneously and demonstrates improved throughput and

fairness. However, the method is effective only for spatially close users and ignores interference and noise effects, limiting real-world applicability.

Gudepu et al. (2024) introduced a drift handling framework for Open RAN consisting of drift detection, analysis, and adaptation stages. The framework improves Quality of Service by adapting AI models when drift is detected. However, the reliance on retraining and pre-certified models introduces computational overhead and may impact near real-time RAN operation.

Samson et al. (2025) proposed a hybrid framework combining LSTM-based traffic prediction and game-theoretic resource allocation for 5G network slicing. While the approach achieves accurate traffic prediction and fair resource allocation, it is evaluated only under simulated environments and assumes simplified network conditions.

Chen et al. (2023) presented RadioSaber, a channel-aware RAN slicing framework with customizable schedulers. The framework improves throughput and reduces flow completion time compared to traditional schedulers. However, the evaluation is limited to downlink scenarios and simulation-based studies.

Polese et al. (2023) demonstrated the feasibility of deploying machine learning-based scheduling xApps in a real Open RAN testbed. Their work highlights the practicality of closed-loop RAN control using the Near-Real-Time RIC. However, the focus is primarily on system integration rather than addressing scheduling drift or long-term adaptability.

Lotfi and Afghah (2023) proposed an O-RAN slicing framework using LSTM-based traffic prediction and deep reinforcement learning for slice management. Although the approach improves resource utilization, it relies on synthetic traffic data and does not address model drift or scalability in real-world deployments.

Orumwense et al. (2023) proposed a rule-based scheduling algorithm for smart grid communications over 5G networks. The approach improves throughput and deadline adherence but lacks learning capability and adaptability to dynamic network conditions.

Elkael et al. (2025) explored the use of large language models to automatically generate RAN schedulers based on high-level intents. While promising, the approach is in its early stages and suffers from reliability and performance limitations compared to expert-driven or learning-based schedulers.

Zhang et al. (2022) proposed a two-phase real-time scheduling framework combining static and adaptive scheduling mechanisms. Although effective in simulations, the heuristic nature of the approach limits robustness under highly dynamic conditions.

Zhang et al. (2022) proposed a "Federated Deep Reinforcement Learning" (FDRL) algorithm for coordinating resource allocation across multiple O-RAN slices. Their approach allows xApps to learn global policies without sharing raw data, enhancing privacy. However, the synchronization of gradients across distributed agents introduces significant latency, making it challenging for sub-millisecond control loops.

Arslan et al. (2023) explored the application of Federated DRL specifically for "Dynamic MAC Scheduling" in O-RAN. By aggregating local models from different base stations, they achieved better generalization. However, the high communication overhead required for frequent model updates can saturate the E2 interface during peak traffic periods.

Lotfi and Afghah (2024) introduced a "Meta-Reinforcement Learning" strategy inspired by Model-Agnostic Meta-Learning (MAML) for adaptive resource optimization. The model is designed to adapt quickly to new network tasks with minimal training steps. However, the computational cost of the meta-update step is prohibitively high for low-power edge devices typical in RAN deployments.

Fiandrino et al. (2023) proposed "EXPLORA," a framework for Explainable AI (XAI) in O-RAN. They argue that black-box DRL controllers often make opaque decisions that network operators cannot trust. While EXPLORA provides feature importance maps to explain scheduling decisions, the added interpretability layer increases inference time, potentially violating the O-RAN real-time latency constraints.

Liang et al. (2024) proposed an intelligent xApp deployment strategy to enhance "Energy Efficiency" in O-RAN. Their method dynamically activates or deactivates network functions based on demand. However, the aggressive power-saving measures often lead to instantaneous throughput drops, negatively impacting User Equipment (UE) with strict Quality of Service (QoS) requirements.

Tsmpazi et al. (2023) conducted a "Comparative Analysis" of various DRL agents (DQN, PPO, A2C) deployed as xApps. Their study provides valuable benchmarking data for O-RAN controllers. However, it is a comparative evaluation that does not propose a novel architectural solution to address the instability of these agents in non-stationary environments.

Sherif et al. (2024) utilized "Transfer Learning" combined with DRL to accelerate resource allocation in O-RAN slicing. By transferring knowledge from a source domain, they reduced training time. However, this approach relies heavily on the similarity between source and target domains; performance degrades significantly when the target environment diverges (negative transfer).

Tuerxun and Nakao (2025) presented "SORA," an online learning framework for energy-efficient resource allocation. SORA continuously updates its policy based on real-time feedback. However, online learning schemes are prone to instability and catastrophic forgetting when exposed to highly volatile and adversarial traffic patterns.

Lozano et al. (2025) introduced "Kairos," a control mechanism for O-RAN Radio Units (O-RUs) that leverages advanced sleep modes to save power. While effective in reducing energy consumption, the wake-up latency of sleep modes introduces jitter, which is unacceptable for Ultra-Reliable Low Latency Communication (URLLC) slices.

Metere et al. (2025) applied "Formal Verification" techniques to ensure service availability and energy efficiency in 6G O-RAN. Their work provides theoretical guarantees for network

behavior. However, the formal models often rely on simplified assumptions that may not hold true in chaotic, real-world wireless channels.

Mudi and Elbiaze (2025) proposed a Federated Reinforcement Learning framework tailored for "Metaverse" applications in O-RAN. Their model optimizes high-bandwidth slicing. However, the extreme sensitivity of Metaverse applications to even minor latency fluctuations poses a challenge that standard RL reward functions struggle to capture effectively.

1.3 Research Gaps

Despite the advancements in 5G scheduling and O-RAN intelligence, several critical gaps remain:

- **Static vs. Dynamic Mismatch:** Most commercial schedulers still rely on static heuristics (like Round Robin or Proportional Fair) that cannot adapt to the rapid temporal fluctuations of modern 5G traffic, leading to inefficient resource utilization.
- **The Retraining Bottleneck:** Existing ML-based schedulers are predominantly supervised, requiring massive labeled datasets and frequent offline retraining. This introduces a significant "blind spot" where the model fails to adapt to new patterns until it is manually retrained and redeployed.
- **Lack of Unsupervised Real-Time Adaptation:** There is a distinct lack of lightweight, unsupervised mechanisms that can detect and correct performance drift *online*, without human intervention or expensive computational overhead.
- **fragmentation of Control Loops:** Current solutions often treat "anomaly detection" and "scheduling control" as separate entities. There is a need for a unified, closed-loop module within the RIC that seamlessly integrates drift detection with immediate policy adaptation.

1.4 Problem Definition

To design and develop a novel drift-aware scheduling algorithm for the Radio Access Network (RAN) system that adaptively optimizes radio resource allocation, aiming to achieve a balance between system throughput, user fairness, and latency when compared to latest scheduling techniques.

1.5 Objectives of the Project

The main objectives of this project are as follows:

- To analyze factors influencing scheduling performance at the MAC layer of 5G RAN systems, focusing on improving system's throughput, user fairness and latency.
- To design and deploy AI/ML models that can learn scheduling patterns and adaptively optimize resource allocation strategies.
- To evaluate the proposed scheduling approach against currently available AI/ML based 5G schedulers, highlighting improvements in system's throughput, user fairness and latency.

Chapter 2

SOFTWARE REQUIREMENT SPECIFICATION

2.1 Overview of SRS

This Software Requirement Specification (SRS) describes the functional and non-functional requirements of the proposed drift-aware scheduling system for 5G O-RAN environments. It defines system behavior, operational constraints, and expected performance.

The system focuses on real-time KPI monitoring, drift detection using a sliding window approach, and adaptive scheduling through a closed-loop control mechanism.

—

2.2 Requirement Specifications

This section outlines the requirements that must be satisfied to ensure correct and efficient system operation under dynamic network conditions.

2.2.1 Functional Requirements

- **FR1 (Data Collection):** The system shall collect real-time Key Performance Indicators (KPIs) such as Throughput, and PRB Utilization from the gNodeB via the O-RAN E2 interface.
- **FR2 (Data Storage):** The system shall persist collected metrics into a time-series data which is written in a .csv file for historical analysis and model validation.
- **FR3 (Drift Detection):** The system shall employ an unsupervised Isolation Forest algorithm to analyze incoming data streams and identify statistical anomalies (concept drift) in real-time.
- **FR4 (Drift Classification):** The system shall distinguish between 'Sudden' (abrupt change) and 'Gradual' (incremental change) drift to determine the severity of the network event.

- **FR5 (Closed-Loop Adaptation):** Upon detecting drift, the system shall automatically trigger a policy based scheduling enforcement (eg: load balancing) via the E2 Control interface.
- **FR6 (Visualization):** The system shall provide a graphical dashboard to display real-time throughput trends, detected drift variance, and the current active scheduling policy.

2.2.2 Non-Functional Requirements

Performance Requirements

- **NFR1 (Latency):** The Drift Detection xApp shall complete the inference cycle (data ingestion to drift score calculation) within **100 ms** to satisfy Near-Real-Time RIC constraints.
- **NFR2 (Throughput):** The system shall support the processing of KPI streams from atmost **20 active UEs** simultaneously without data loss.
- **NFR3 (Resource Efficiency):** The machine learning inference engine shall consume no more than **30% of available CPU/GPU resources** on the RIC host to ensure coexistence with other xApps.

Reliability and Availability

- **NFR4 (Availability):** The scheduling module shall maintain **99.9% availability** during the experimental timeframe, handling network jitter without crashing.
- **NFR5 (Recovery Time):** In the event of an xApp crash or E2 link failure, the system shall revert to the default static scheduler within **2 seconds** to prevent service outage.

Scalability and Usability

- **NFR6 (Scalability):** The .csv file shall handle write speeds of up to **1000 points/second**, allowing for future network densification.
- **NFR7 (Responsiveness):** The Visualization Dashboard shall update system metrics with a latency satisfying the **soft real-time** constraint (less than 1 second refresh rate).

Chapter 3

MATHEMATICAL MODELING

3.1 Overview

The proposed Drift-Aware Adaptive Scheduling system employs a mathematical model to balance network stability and performance. The core components include an optimization objective function, system constraints, and an anomaly detection model based on Isolation Forest.

3.2 Objective Function

The optimization objective is to minimize drift-induced abnormal behavior while maximizing system throughput across evaluation windows. The objective function J is defined as:

$$\min(J) = \frac{1}{T} \sum_{t=1}^T (\alpha \cdot \mathcal{D}_t - \beta \cdot \text{Throughput}_t) \quad (3.1)$$

Where:

- T : Total number of evaluation windows (time intervals).
- $\mathcal{D}_t$: Drift Score at time t , derived from the Isolation Forest model.
- Throughput_t : Measured data transmission rate at time t .
- α, β : Weighting factors that control the trade-off between model precision (stability) and throughput optimization.

The weighting factors α and β are adjusted based on network conditions:

- **Case 1 (Stable Throughput):** When the network is stable, the focus is on stability. Thus, $\alpha > \beta$.
- **Case 2 (Sudden Throughput Change):** When throughput fluctuates significantly or drift scores rise abnormally, the focus shifts to performance. Thus, $\beta > \alpha$.

3.3 System Constraints

To ensure Quality of Service (QoS) and respect physical resource limitations, the system operates under the following constraints:

3.3.1 Minimum Throughput Constraint

Each User Equipment (UE) u must receive at least a minimum required throughput ($R_{\min}$) to maintain service continuity:

$$\text{Throughput}_u \geq R_{\min}, \quad \forall u \in \text{Active UEs} \quad (3.2)$$

In this system, the minimum throughput is considered 0 Mbps (best effort) and maximum is 20 Mbps, with a baseline reference of 10 Mbps.

3.3.2 Physical Resource Block (PRB) Constraint

The allocation of Physical Resource Blocks (PRBs) must not exceed the total available bandwidth resources. For a 10 MHz channel bandwidth with 15 kHz subcarrier spacing, the total available PRBs ($N_{\text{PRB}}^{\text{total}}$) is 52.

$$\sum_u \text{PRB}_u \leq N_{\text{PRB}}^{\text{total}} \quad (3.3)$$

Where $5 \leq N_{\text{PRB}}^{\text{total}} \leq 52$.

3.4 Isolation Forest Anomaly Detection Model

The system employs an unsupervised Isolation Forest algorithm to detect deviations in network behavior.

3.4.1 Feature Vector

The network state at time t is represented by a feature vector $\mathbf{x}_t$, consisting of d Key Performance Indicators (KPIs):

$$\mathbf{x}_t = [k_t^1, k_t^2, \dots, k_t^d] \quad (3.4)$$

Specifically, $\mathbf{x}_t = [\text{DL Throughput}, \text{UL Throughput}, \text{DL PRB}, \text{UL PRB}]$.

3.4.2 Anomaly Scoring

The Isolation Forest algorithm isolates observations by randomly selecting a feature and then randomly selecting a split value. The anomaly score $s(\mathbf{x}_t)$ is defined as:

$$s(\mathbf{x}_t) = 2^{-\frac{E(h(\mathbf{x}_t))}{c(n)}} \quad (3.5)$$

Where:

- $h(\mathbf{x}_t)$: Path length, representing the number of edges $\mathbf{x}_t$ traverses from the root to an external node.
- $E(h(\mathbf{x}_t))$: Average path length across all isolation trees.
- $c(n)$: Average path length of unsuccessful search in a Binary Search Tree (BST), used for normalization, given by:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n} \quad (3.6)$$

Here, $H(i)$ is the harmonic number estimated as $\ln(i) + 0.5772156649$ (Euler's constant).

The score $s(\mathbf{x}_t) \in [0, 1]$ is interpreted as:

- $s \approx 1$: Highly anomalous (Abnormal behavior).
- $s \approx 0.5$: No distinct anomaly (BS line).
- $s \approx 0$: Normal behavior.

3.5 Drift Scoring and Classification

Instantaneous anomaly scores are smoothed using an Exponential Moving Average (EMA) to calculate the Drift Score $\mathcal{D}_t$, minimizing noise:

$$\mathcal{D}_t = \lambda \cdot s(\mathbf{x}_t) + (1 - \lambda) \cdot \mathcal{D}_{t-1} \quad (3.7)$$

Where $\lambda \in (0, 1)$ is the smoothing factor controlling sensitivity.

3.5.1 Drift Threshold Criteria

Drift is classified based on the magnitude and rate of change of $\mathcal{D}_t$ relative to a threshold θ and rate parameter k :

Table 3.1: Drift Classification Criteria

Drift Type	Condition	Interpretation
Stable (No Drift)	$\mathcal{D}_t \leq \theta$	Normal network behavior
Incremental Drift	$\mathcal{D}_t > \theta$ and $ \mathcal{D}_t - \mathcal{D}_{t-1} < k$	Gradual degradation over time
Sudden Drift	$\mathcal{D}_t > \theta$ and $ \mathcal{D}_t - \mathcal{D}_{t-1} > k$	Abrupt change (e.g., UE disconnection)

3.6 Scheduler Control Decisions

Based on the classified drift type, the RIC enforces specific scheduling actions:

Table 3.2: Scheduler Control Decisions

Drift Type	Scheduler Action
Stable	Continue normal scheduling (Monitor)
Incremental	Gradual parameter tuning (e.g., Load Balancing)
Sudden	Immediate Retraining

Chapter 4

SYSTEM DESIGN

The design of the proposed system leverages the modularity and intelligence of the O-RAN architecture to address the limitations of static 5G scheduling. This chapter details the comprehensive system architecture, delineating the interaction between the data acquisition, intelligence, and control layers.

The core philosophy of the design is closed-loop feedback: the system continuously senses the radio environment, identifies performance drift via unsupervised learning, and autonomously actuates scheduling policies to correct deviations. By decoupling the control logic (xApp) from the hardware (gNodeB), the system achieves a level of flexibility and real-time responsiveness that is unattainable in traditional monolithic RAN architectures. The subsequent sections provide a granular breakdown of the specific modules—from the RAN data telemetry to the Isolation Forest-based inference engine—that collectively enable this adaptive capability.

4.1 Architecture Design

The system architecture is designed based on the O-RAN module and consists of three main components: Flexible RIC, KPI Monitor xApp, and Drift Detection Engine.

The major components of the system are:

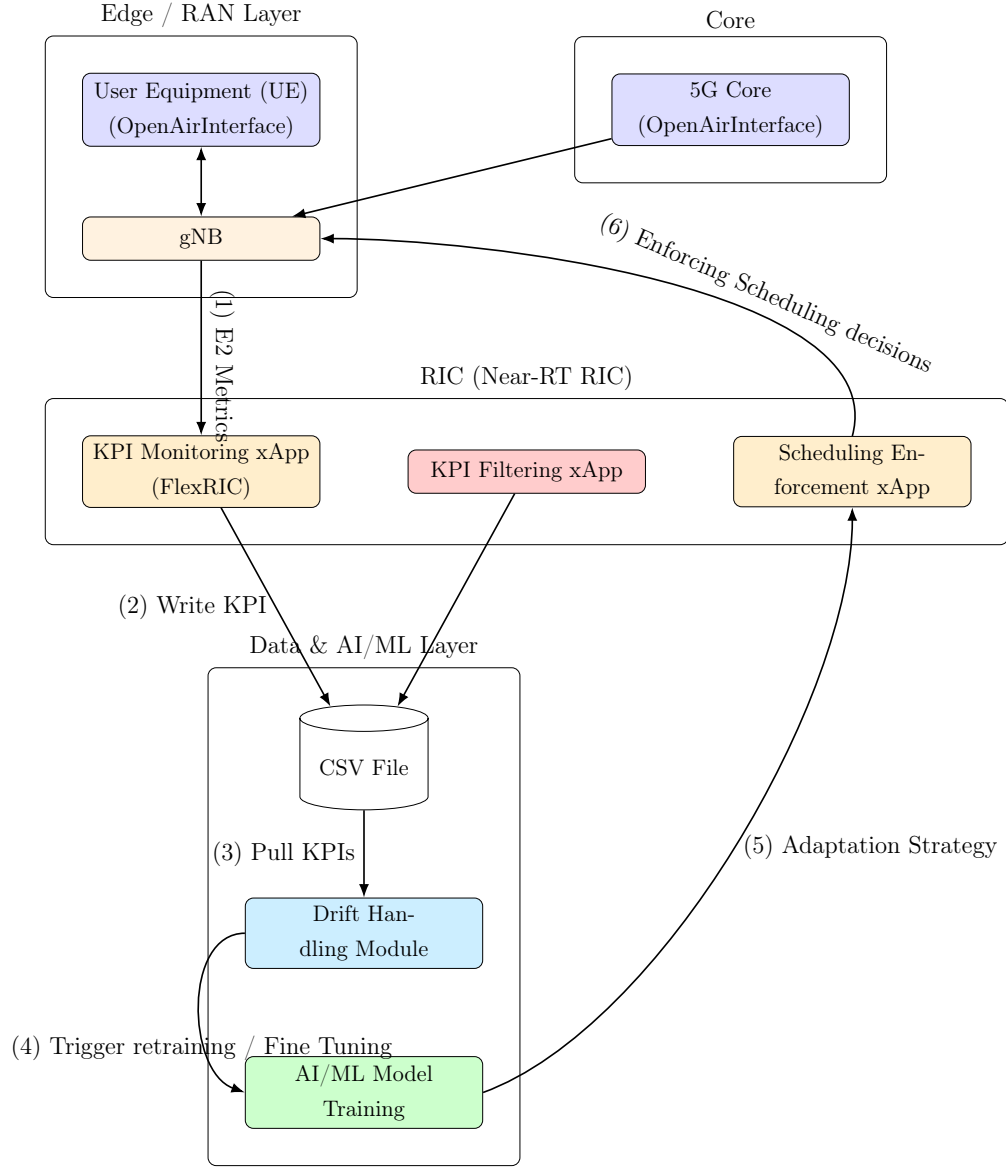


Figure 4.1: Overall Architecture of the Proposed Drift-Aware Scheduling System

The architecture illustrated in 4.1 represents a comprehensive O-RAN-compliant system designed for intelligent closed-loop control. At the physical layer, the gNodeB exposes internal state metrics through the standardized E2 Interface, connecting the RAN to the Flexible RIC (FlexRIC). This interface serves as the bidirectional conduit for both telemetry data flowing upward and control policies flowing downward. Inside the RIC, the KPI Monitor xApp acts as the data ingress pipeline, sanitizing raw metric streams, while the Drift Detection Engine serves as the intelligence core, processing these streams via unsupervised learning to identify anomalies. The Scheduling Enforcement xApp then translates high-level decisions into concrete MAC-layer commands. This separation of concerns ensures that the heavy computational load of ML inference does not block the real-time scheduling operations of the gNodeB, while the connection to the 5G Core Network maintains end-to-end service continuity.

4.2 Workflow of the Proposed System

The proposed system operates in a continuous closed-loop manner to monitor and adapt scheduling decisions in real time.

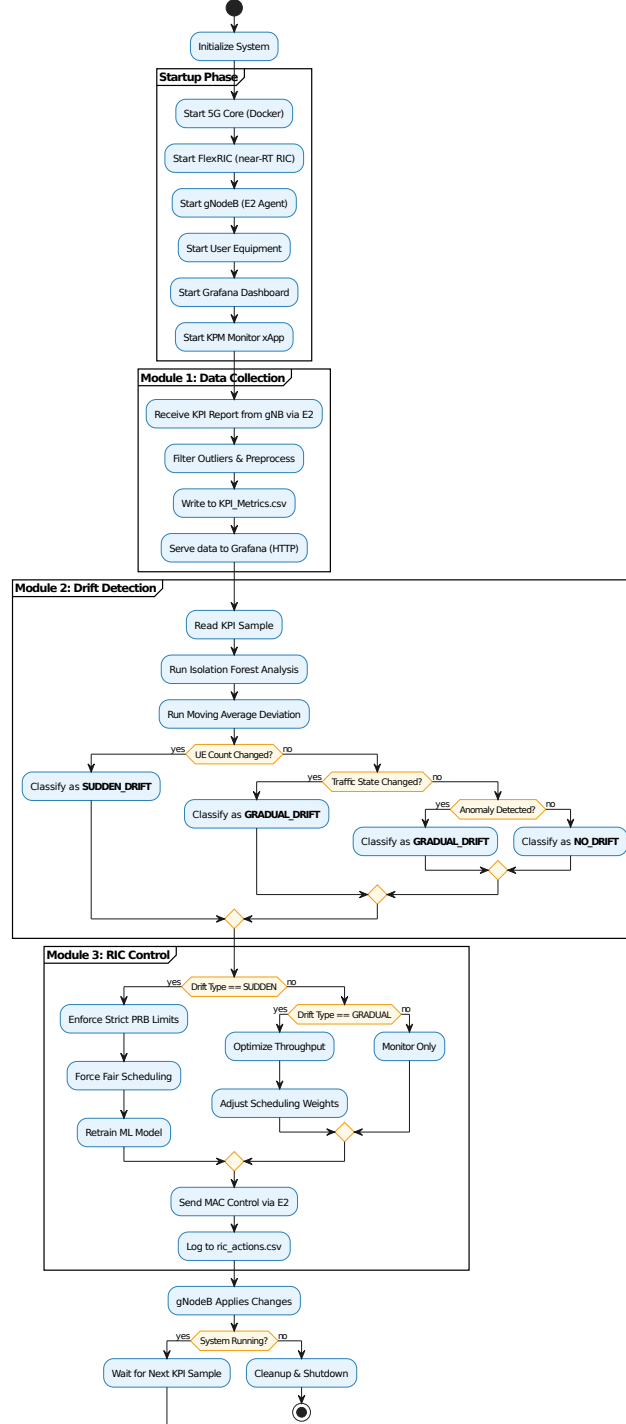


Figure 4.2: Closed-Loop Workflow of Drift Detection and Scheduling Adaptation

Figure 4.2 illustrates the complete operational workflow of the O-RAN testbed, beginning with system initialization and continuing through a continuous closed-loop automation cycle.

During the **startup phase**, the system sequentially launches the 5G Core, FlexRIC, gNodeB, User Equipment (UE), Grafana Dashboard, and the KPM Monitor xApp.

Module 1 (Data Collection) receives Key Performance Indicator (KPI) reports from the gNodeB via the E2 interface. The collected data undergoes outlier filtering, after which the cleaned KPI samples are written to CSV files and simultaneously served to the Grafana dashboard for real-time visualization.

Module 2 (Drift Detection) processes each incoming KPI sample using a combination of Isolation Forest and Moving Average algorithms. Based on the observed network behavior, the system classifies its operational state as follows: changes in UE count are identified as **SUDDEN_DRIFT**, traffic state transitions are categorized as **GRADUAL_DRIFT**, detected KPI anomalies are also treated as **GRADUAL_DRIFT**, while stable network conditions are labeled as **NO_DRIFT**.

Module 3 (RIC Control) executes policy-driven control actions according to the detected drift type. A **SUDDEN_DRIFT** event triggers strict Physical Resource Block (PRB) enforcement and model retraining. In the case of **GRADUAL_DRIFT**, the system performs throughput optimization and scheduling adjustments. When no drift is detected (**NO_DRIFT**), the system remains in monitoring mode. All selected actions are transmitted to the gNodeB as MAC control messages via the E2 interface and logged in the `ric_actions.csv` file.

After the gNodeB applies the received control actions, the system waits for the next KPI sample, thereby forming a closed-loop feedback mechanism that continuously adapts RAN resource allocation to changing network conditions until system shutdown.

4.2.1 KPI Monitor xApp

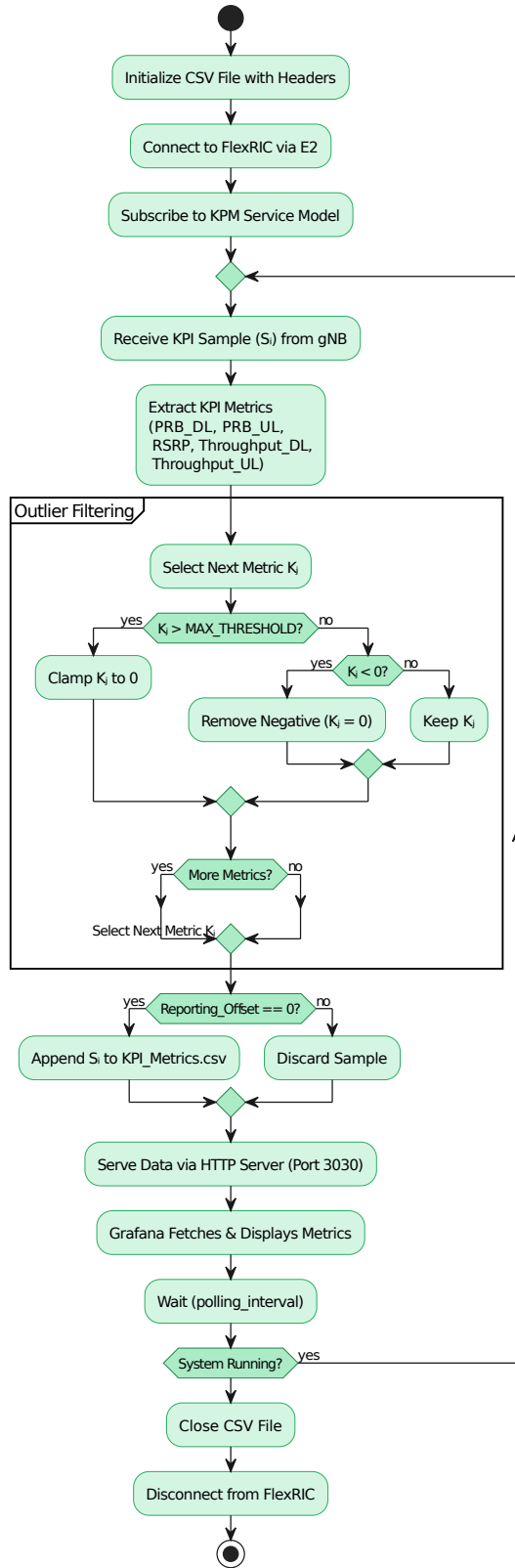


Figure 4.3: Workflow of KPI Monitor xApp

The flowchart 4.3 depicts the continuous KPI data collection and preprocessing pipeline that supplies clean, validated metrics to the drift detection subsystem.

The process begins with system initialization, during which the CSV file is created with appropriate headers, an E2 connection to FlexRIC is established, and a subscription to the KPM Service Model is registered. The pipeline then enters its main processing loop, where KPI samples are received from the gNodeB. Each sample includes Physical Resource Block (PRB) allocation for downlink and uplink, Reference Signal Received Power (RSRP) measurements, and bidirectional throughput statistics.

Upon reception, each KPI sample undergoes an outlier filtering stage. Individual metrics are validated against predefined constraints: values exceeding a `MAX_THRESHOLD` of 1,000,000 are clamped to zero to prevent overflow-induced corruption, while negative values are discarded as physically invalid measurements.

Following validation, the system verifies the sample's `Reporting_Offset`. Only samples with an offset value of zero are appended to the `KPI_Metrics.csv` file to ensure temporal consistency, whereas samples with non-zero offsets are discarded.

The validated KPI data is subsequently served through an HTTP server operating on port 3030, enabling real-time visualization in Grafana dashboards. These dashboards display throughput trends, PRB utilization charts, and active UE counts.

After data publication, the pipeline waits for the configured polling interval (typically one second) before processing the next KPI sample, thereby generating a continuous time-series data stream. This preprocessing pipeline enforces data quality by eliminating outliers and invalid measurements prior to persistence, preventing corruption in downstream analytics.

The process continues indefinitely until system shutdown, at which point the CSV file is closed and the FlexRIC connection is gracefully terminated.

Algorithm 1: KPI Data Collection and Filtering

Require: Raw KPI reports from gNB via E2 interface, `MAX_THRESHOLD` (1,000,000)

Ensure: Cleaned KPI data written to CSV file

Variables: S_i (i^{th} sample), K_j (j^{th} metric), `CSV_FILE`

Initialize `CSV_FILE` with headers

Connect to FlexRIC via E2 interface

Subscribe to KPM Service Model

loop

 {Runs every polling interval (1s)}

 Receive S_i from gNB


```
for each  $K_j$  in  $S_i$  do
  if  $K_j > \text{MAX\_THRESHOLD}$  then
     $K_j \leftarrow 0$  {Clamp outlier}
  end if
  if  $K_j < 0$  then
     $K_j \leftarrow 0$  {Remove negative values}
  end if
end for
if  $S_i.\text{Reporting\_Offset} == 0$  then
  Append  $S_i$  to CSV_FILE
end if
Send  $S_i$  to HTTP server for Grafana
end loop
```

4.2.2 Flexible RIC Module

The Flexible RIC enables near real-time control of RAN functions. It hosts xApps that facilitate closed-loop scheduling adaptation by enforcing control decisions on the gNodeB via the E2 interface.

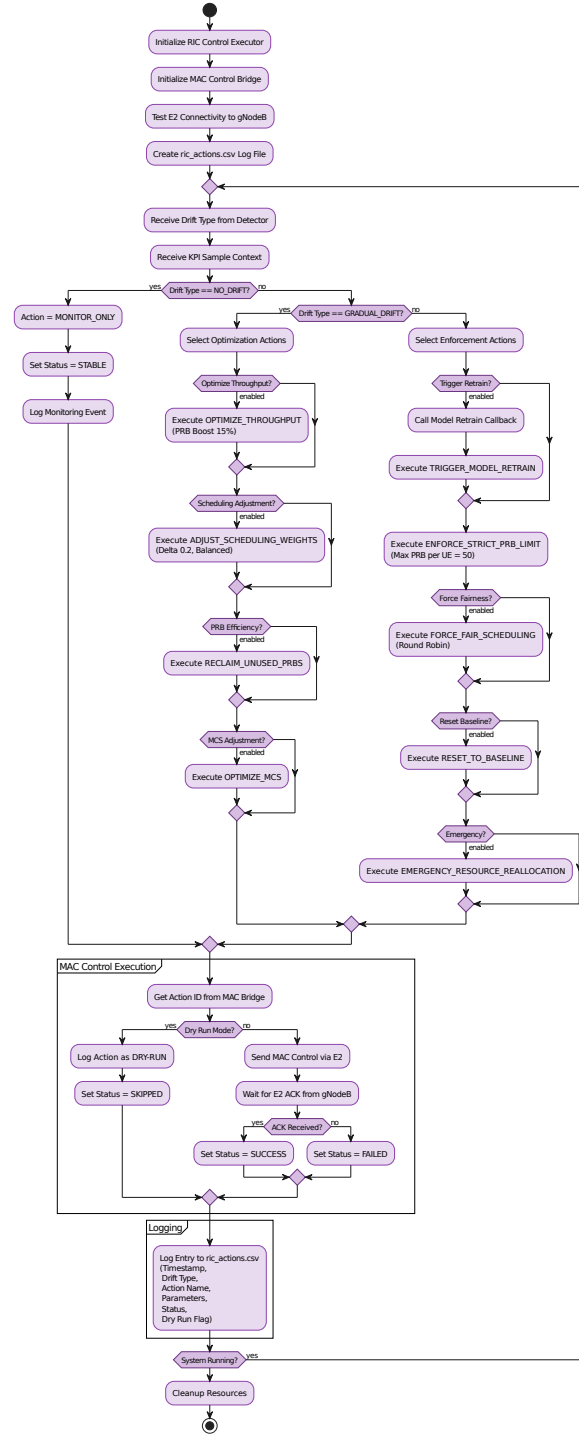


Figure 4.4: Workflow of Flexible RIC

The flowchart 4.4 illustrates a dual-method anomaly detection workflow that combines supervised machine learning with statistical analysis to classify system drift.

The **Training Phase** begins by loading historical KPI data from CSV files and filtering samples with `Reporting_Offset = 0`. The dataset is then partitioned into traffic and no-traffic observations to form a balanced training set, consisting of 80% traffic samples and

20% no-traffic samples. Feature vectors are normalized using a *StandardScaler*, after which an *Isolation Forest* model is trained with 100 estimators and a contamination rate of 10%. An anomaly threshold is established at the 10th percentile of the resulting training anomaly scores.

During the **Detection Loop**, each incoming KPI sample undergoes parallel analysis. First, the Isolation Forest computes anomaly scores on the scaled feature vectors and flags samples whose scores fall below the learned threshold. In parallel, a *Moving Average Deviation* method evaluates statistical deviations from rolling windows of 30 samples, triggering anomaly alerts when metric deviations exceed a three-sigma threshold.

Based on the outputs of both detection mechanisms, the system classifies anomaly severity into four levels: **HIGH** severity when both methods concur on the presence of an anomaly, **MEDIUM** severity when either method detects anomalous behavior, **LOW** severity for borderline signals, and **NORMAL** for stable operation.

Drift Classification is performed using hierarchical decision logic that maintains **UE_History** and **Traffic_History** queues. Abrupt changes in UE count are classified as **SUDDEN_DRIFT**, indicating a critical topology change (red). Traffic start or stop events, as well as medium- or high-severity anomalies, are categorized as **GRADUAL_DRIFT**, reflecting evolving traffic patterns (yellow). In the absence of significant deviations, the system reports **NO_DRIFT**, corresponding to stable network operation (green).

The classified drift type is forwarded to the RIC Controller for subsequent action selection. This continuous closed-loop process enables real-time network state monitoring with sub-second responsiveness to anomalous conditions.

Algorithm 2: Flexible RIC Execution

Require: Drift_Type, S_i , Config

Ensure: Control actions executed via FlexRIC E2 interface

Variables: MAC_Bridge, Action_List

Load RIC control configuration; Initialize MAC_Bridge

PHASE B: Action Selection

if Drift_Type == NO_DRIFT **then**

 Action $\leftarrow$ "MONITOR_ONLY"

else if Drift_Type == GRADUAL_DRIFT **then**

if Config.optimize_throughput **then**

 Execute "OPTIMIZE_THROUGHPUT"

end if

```

[Other gradual optimizations...]
else if Drift_Type == SUDDEN_DRIFT then
  if Config.trigger_retrain then
    Execute "TRIGGER_MODEL_RETRAIN"
  end if
  Execute "ENFORCE_STRICT_PRB_LIMIT"
  [Other sudden measures...]
end if
PHASE C: MAC Control Execution
for each Action in Action_List do
  Send MAC control via E2 Interface
  Wait for ACK
end for
Log actions to ric_actions.csv

```

4.2.3 Class Diagram

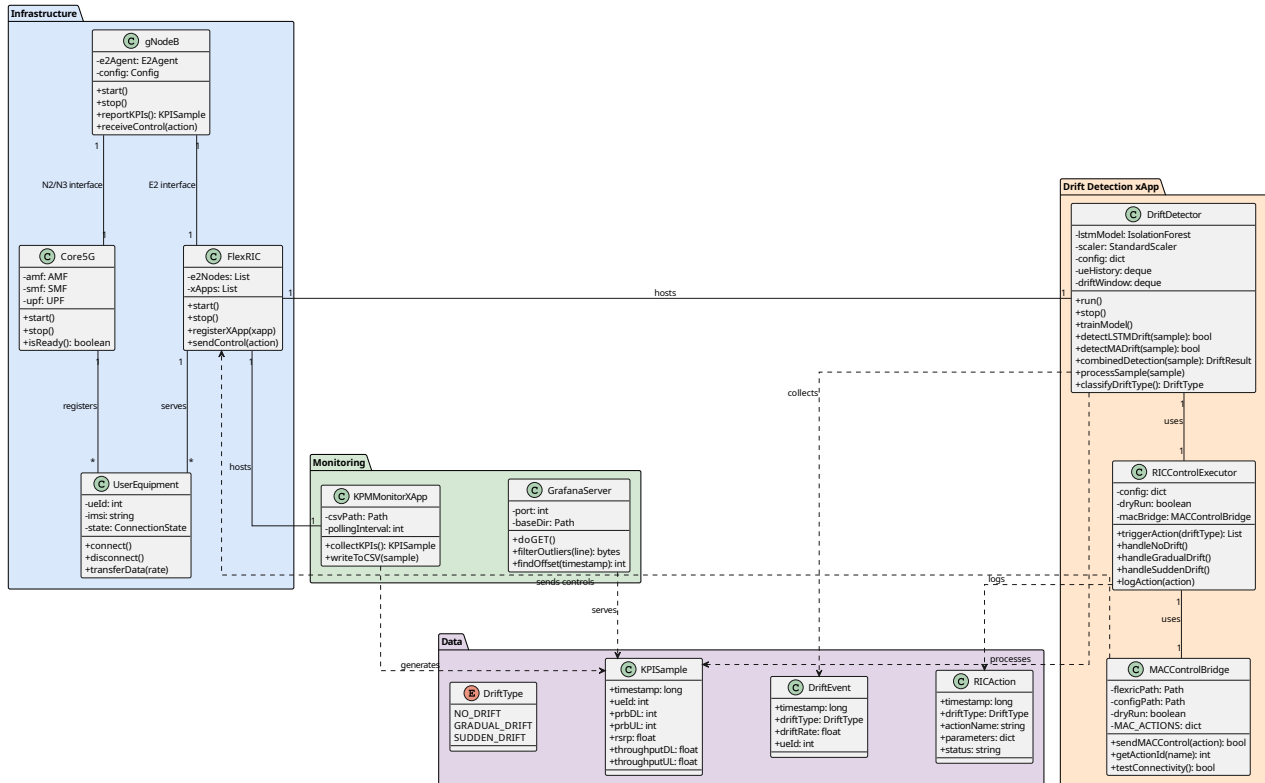


Figure 4.5: Class Diagram of the Drift-Aware Scheduling System

Figure 4.7 illustrates a four-layer, object-oriented architecture implementing closed-loop automation for intelligent RAN resource management using machine learning-based drift detection.

Layer 1 (Infrastructure – Blue) comprises the foundational 5G components, including *Core5G* (AMF, SMF, and UPF responsible for user registration, session management, and routing), the *gNodeB* (base station equipped with an E2 Agent for KPI reporting and MAC-layer control), *UserEquipment* (mobile devices identified by IMSI/UE IDs), and *FlexRIC*, which serves as the near-RT RIC orchestration platform hosting xApps.

Layer 2 (Drift Detection xApp – Orange) implements the core intelligence of the system. It consists of the *DriftDetector*, which integrates an Isolation Forest with a moving-average engine to classify drift as `NO_DRIFT`, `GRADUAL_DRIFT`, or `SUDDEN_DRIFT`. The *RICControlExecutor* maps detected drift events to policy-based actions, enabling monitoring, optimization, or enforcement strategies. These actions are translated into MAC-layer commands through the *MACControlBridge*, which interfaces with FlexRIC using E2 MAC Service Model identifiers.

Layer 3 (Monitoring – Green) provides system observability. The *KPMMonitorXApp* collects key performance metrics such as PRB utilization, RSRP, and throughput, exporting them to CSV format. Visualization is supported by the *GrafanaServer*, an HTTP server operating on port 3030, which applies outlier filtering prior to dashboard rendering.

Layer 4 (Data Model – Purple) defines the data structures exchanged across layers. These include *KPISample* (a snapshot of network measurements), *DriftEvent* (an anomaly record enriched with contextual information), *RICAction* (a log of executed control commands), and the *DriftType* enumeration.

Key relationships within the architecture include *composition* (the *DriftDetector* owns the *RICControlExecutor*, which in turn owns the *MACControlBridge*), *aggregation* (FlexRIC hosting multiple xApps), *association* (the *gNodeB* interfacing with FlexRIC via the E2 interface and with Core5G via N2/N3), and *dependency* relationships represented by data flow through *KPISample*, *DriftEvent*, and *RICAction* objects.

Overall, the architecture leverages the *Strategy*, *Observer*, and *Bridge* design patterns to enable modular, extensible, and automated RAN resource management under dynamic network conditions.

4.2.4 Drift Detection Engine

The drift detection engine uses an unsupervised Isolation Forest model operating on sliding windows of KPI data. Each window represents recent network behavior. Higher anomaly scores indicate abnormal conditions, which are interpreted as performance drift. Temporal analysis allows classification of drift as sudden or gradual.

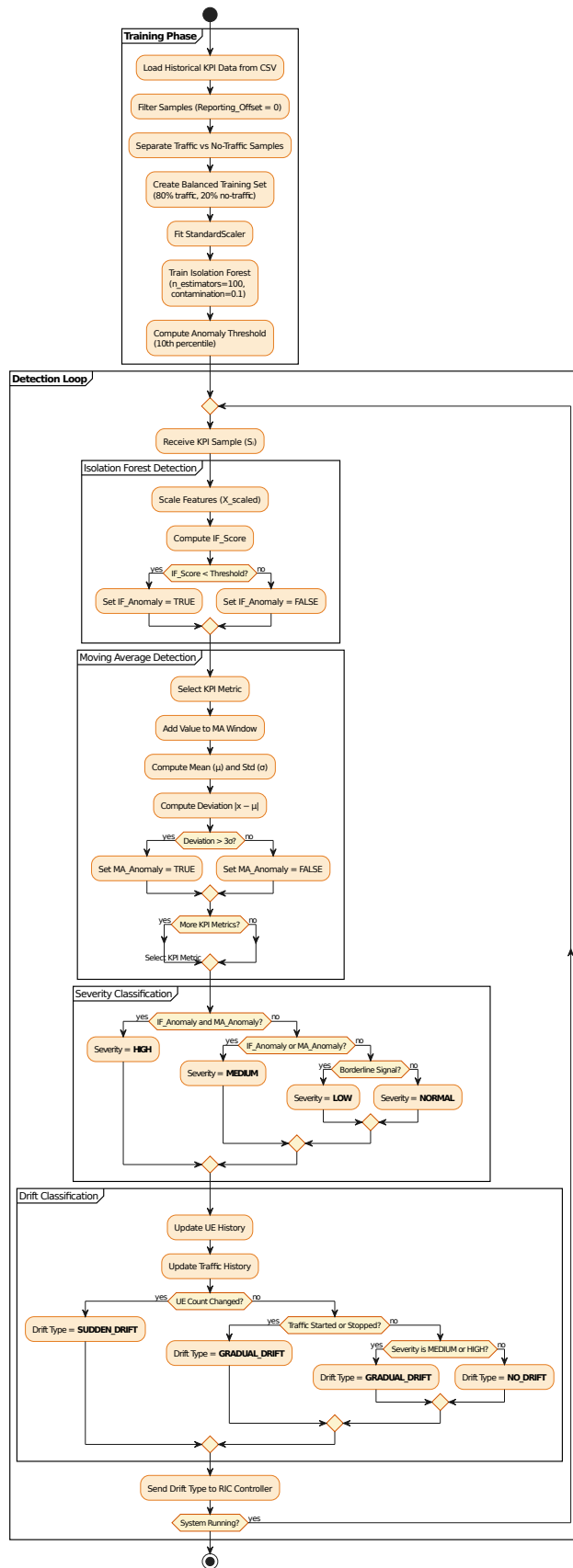


Figure 4.6: Workflow of Drift Detection Engine

The flowchart 4.6 illustrates the policy-based control action execution workflow that translates detected drift events into RAN resource management commands.

The process begins with system initialization, during which the *RIC Control Executor* and *MAC Control Bridge* are instantiated, E2 connectivity is verified, and the `ric_actions.csv` log file is created. The system then enters its main control loop, where it receives `Drift_Type` classifications along with `KPI_Sample` context information from the Drift Detector.

Control logic is implemented using a three-way decision structure based on drift severity. When `NO_DRIFT` is detected, the system operates in monitoring-only mode, performing no control intervention and logging the stable network state.

For `GRADUAL_DRIFT` events, the controller selectively executes optimization actions according to predefined configuration flags. These actions may include `OPTIMIZE_THROUGHPUT`, which applies a 15% PRB boost to high-traffic UEs; `ADJUST_SCHEDULING_WEIGHTS`, which applies a scheduling weight delta of 0.2 to achieve balanced resource allocation; `RECLAIM_UNUSED_PRBS`, which redistributes underutilized resources; and `OPTIMIZE_MCS`, which adapts modulation and coding schemes based on channel conditions.

In the case of `SUDDEN_DRIFT`, the system enforces strict control measures. These include triggering model retraining via callback, executing `ENFORCE_STRICT_PRB_LIMIT` to impose a hard cap of 50 PRBs per UE, activating `FORCE_FAIR_SCHEDULING` using round-robin allocation with equal weights, performing `RESET_TO_BASELINE` to restore default configuration parameters, and initiating `EMERGENCY_RESOURCE_REALLOC` to redistribute all available resources with fairness priority.

During the **MAC Control Execution** phase, selected actions are translated into MAC Service Model action identifiers. Depending on the execution mode, the system either logs dry-run operations or transmits E2 Control Requests to the gNodeB and waits for acknowledgment. All executed actions are recorded in the `ric_actions.csv` file, including timestamps, drift types, action names, parameter values, execution status, and dry-run indicators to support auditing and traceability.

The control loop continues indefinitely until system shutdown, thereby implementing a continuous closed-loop RAN control mechanism.

Algorithm 3: Drift Detection (Isolation Forest + Moving Average)

Require: S_i (KPI sample), Historical KPI data

Ensure: Drift Type: {`NO_DRIFT`, `GRADUAL_DRIFT`, `SUDDEN_DRIFT`}

Variables: IF_Model (ISO Forest), Scaler, MA_Window, Drift_Threshold

PHASE A: Model Training

Load historical KPI data; Filter Reporting_Offset = 0
 Create balanced training set and extract feature vector X
 Fit Scaler and Train IF_Model; Compute Drift_Threshold

PHASE B: Real-Time Drift Detection

```

for each new sample  $S_i$  do
   $X \leftarrow$  Extract features from  $S_i$ ;  $X_{scaled} \leftarrow$  Scaler.transform( $X$ )
  IF_Score  $\leftarrow$  IF_Model.score_samples( $X_{scaled}$ )
  if IF_Score < Drift_Threshold then
    IF_Drift  $\leftarrow$  TRUE
  else
    IF_Drift  $\leftarrow$  FALSE
  end if
  for each metric  $K_j$  in  $S_i$  do
    Append  $K_j$  to MA_Window[ $K_j$ ]
    if deviation >  $3\sigma$  then
      MA_Drift  $\leftarrow$  TRUE
    end if
  end for
  Determine Severity (HIGH/MEDIUM/LOW/NORMAL) based on drifts
PHASE C: Drift Classification
  Detect UE changes and Traffic State changes
  if UE_Changed then
    return SUDDEN_DRIFT
  else if Traffic_Stopped OR Traffic_Started OR Severity  $\geq$  MEDIUM then
    return GRADUAL_DRIFT
  else
    return NO_DRIFT
  end if
end for

```

4.2.5 Class Diagram

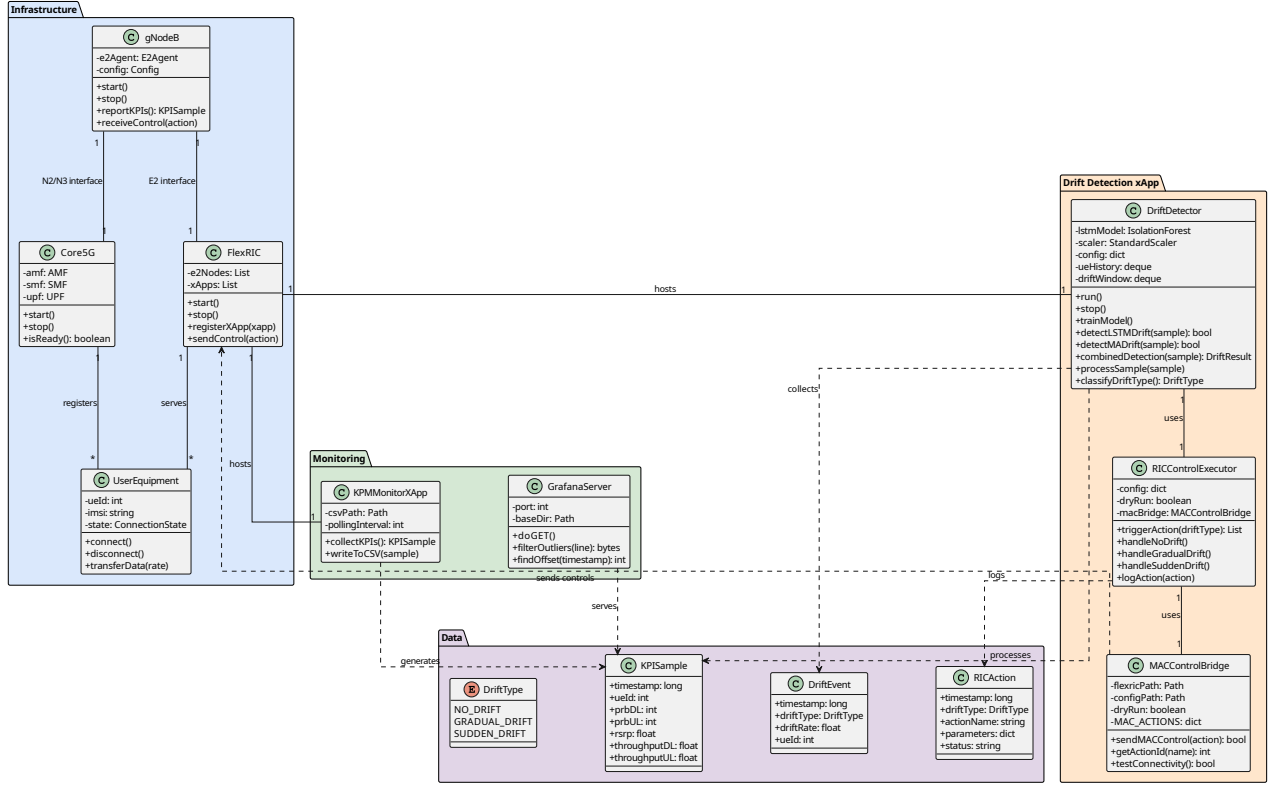


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MACControlBridge, which interfaces with FlexRIC using E2 MAC Service Model identifiers.

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Overall, the architecture leverages the *Strategy*, *Observer*, and *Bridge* design patterns to enable modular, extensible, and automated RAN resource management under dynamic network conditions.

4.2.6 Sequence Diagram

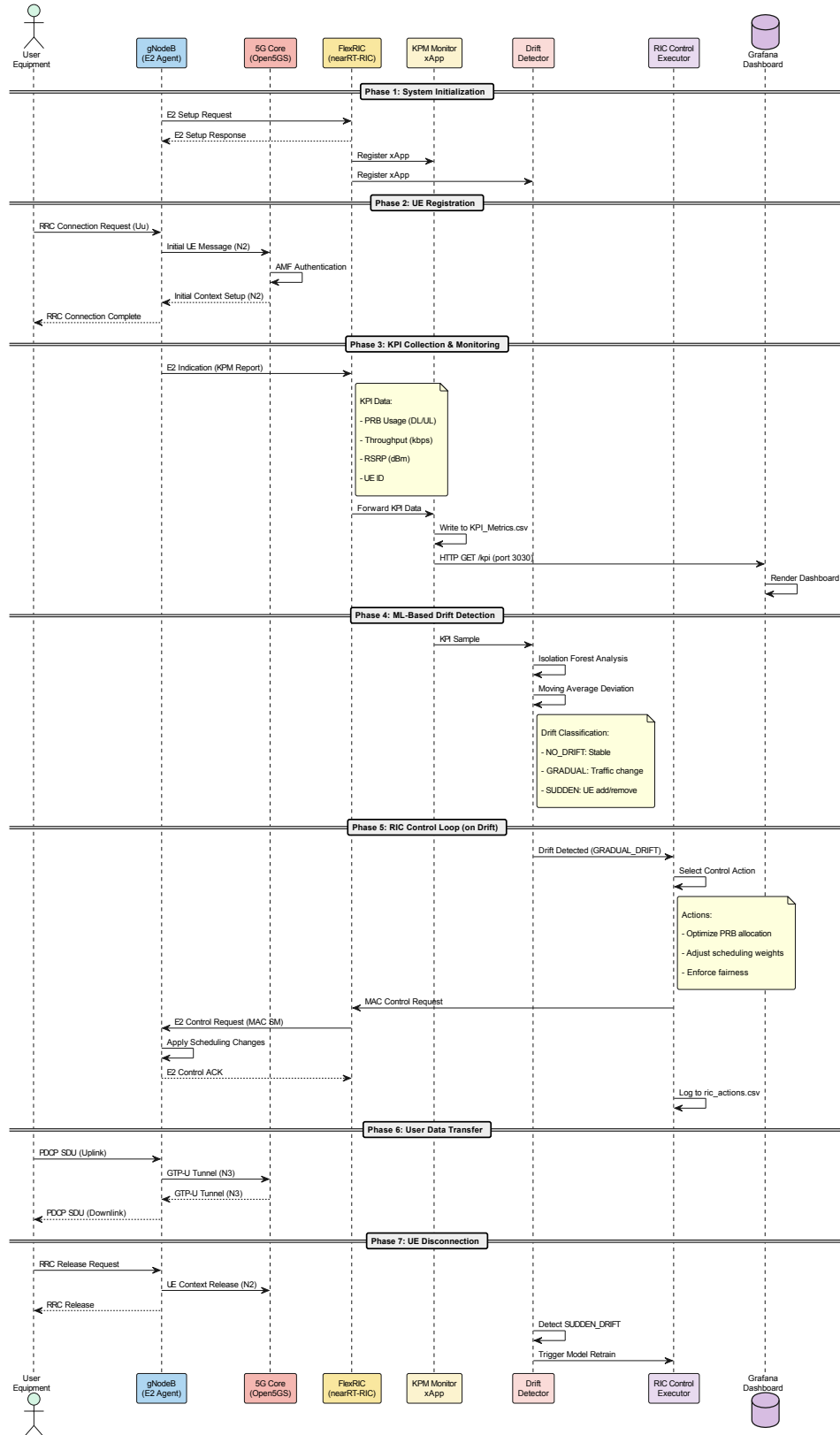


Figure 4.8: Sequence Diagram of Drift Detection and Scheduling Adaptation

Figure 4.8 illustrates the temporal message flow and interactions among eight key system components across seven operational phases.

Phase 1 (System Initialization) establishes the foundational control-plane connectivity. The gNodeB initiates the process by sending an *E2 Setup Request* to FlexRIC, which responds with an acknowledgment. Subsequently, FlexRIC registers the KPM Monitor xApp and the Drift Detector xApp, enabling KPI monitoring and intelligent control functionalities.

Phase 2 (UE Registration) depicts the access and authentication procedure. The User Equipment (UE) initiates an *RRC Connection Request* over the Uu interface to the gNodeB. The gNodeB forwards an *Initial UE Message* to the 5G Core via the N2 interface, triggering authentication procedures at the AMF. Upon successful authentication, the Core returns an *Initial Context Setup* message, after which the UE completes the process with an *RRC Connection Complete* message.

Phase 3 (KPI Collection and Monitoring) represents the continuous monitoring loop. The gNodeB periodically transmits *E2 Indication* messages containing KPI metrics such as PRB utilization, throughput, RSRP, and UE identifiers to FlexRIC. These indications are forwarded to the KPM Monitor xApp, which persists the data in CSV format and serves it via an HTTP interface to Grafana for real-time dashboard visualization.

Phase 4 (ML-Based Drift Detection) illustrates the analytical pipeline. The KPM Monitor forwards validated KPI samples to the Drift Detector xApp, where dual-method analysis is performed using Isolation Forest and Moving Average algorithms. Based on the detected patterns, the system classifies the operational state as **NO_DRIFT** (stable conditions), **GRADUAL_DRIFT** (traffic pattern changes), or **SUDDEN_DRIFT** (topology or UE count changes).

Phase 5 (RIC Control Loop) is activated upon drift detection. The Drift Detector sends the classified drift type to the RIC Controller, which selects appropriate control actions. For example, a **GRADUAL_DRIFT** event triggers PRB optimization policies. The selected actions are transmitted as *E2 Control Requests* through FlexRIC to the gNodeB using MAC Service Model primitives. Upon receiving acknowledgments from the gNodeB, all executed actions are logged in the `ric_actions.csv` file.

Phase 6 (User Data Transfer) demonstrates the user-plane operation. The UE transmits PDCP Service Data Units (SDUs) in the uplink to the gNodeB, which encapsulates the data into GTP-U tunnels and forwards it to the 5G Core via the N3 interface. The downlink data path follows the reverse process, delivering user data from the Core to the UE.

Phase 7 (UE Disconnection) illustrates graceful session termination. The UE initiates an *RRC Release Request*, after which the gNodeB signals a *UE Context Release* to the 5G Core via the N2 interface. Upon confirmation, the gNodeB completes the process by sending an *RRC Release* message to the UE. This topology change is identified by the Drift Detector as a **SUDDEN_DRIFT** event, triggering model retraining to adapt the system to the updated network state.

Chapter 5

IMPLEMENTATION

5.1 Overview

This chapter details the implementation of the proposed Drift-Aware Scheduling system using the O-RAN software stack. The implementation is divided into two phases: first, understanding the baseline Proportional Fair scheduler used in the default OpenAirInterface (OAI) gNodeB, and second, developing the intelligent O-RAN xApp that modifies this behavior in real-time based on concept drift.

5.2 Default gNodeB MAC Layer Scheduler

The default scheduler in OpenAirInterface (OAI) operates on a standard Proportional Fair (PF) algorithm. The PF scheduler aims to balance system throughput and user fairness by assigning resources to users with good channel conditions while ensuring that users with poor conditions are not starved.

The core metric calculated for each User Equipment (UE) k on Resource Block n at time t is:

$$M_{k,n}(t) = \frac{R_{k,n}(t)}{\bar{R}_k(t)} \quad (5.1)$$

Where:

- $R_{k,n}(t)$: Estimated instantaneous data rate on RB n .
- $\bar{R}_k(t)$: Average past throughput of user k .

A representative code snippet of the default OAI scheduler logic (simplified for clarity) is shown below. This static logic runs locally on the gNodeB and does not account for external drift or global traffic patterns.

Algorithm 4: Default Proportional Fair Metric Calculation (C++)

```

1 // OAI Default Scheduler: nr_schedule_ue_spec.c
2 void pf_scheduler(module_id_t module_id, frame_t frame, sub_frame_t slot) {
3     for (int ue_id = 0; ue_id < active_ues; ue_id++) {
4         // Calculate instantaneous rate (MCS based)
5         float instantaneous_rate = get_estimated_rate(ue_id);
6
7         // Retrieve moving average throughput
8         float avg_throughput = ue_stats[ue_id].avg_throughput;
9
10        // Calculate Proportional Fair metric
11        float pf_metric = instantaneous_rate / (avg_throughput + 1e-9);
12
13        // Assign RB to UE with highest metric
14        if (pf_metric > max_metric) {
15            max_metric = pf_metric;
16            selected_ue = ue_id;
17        }
18    }
19    allocate_resources(selected_ue);
20 }

```

5.3 Drift-Aware Scheduler Implementation

To address the limitations of the static scheduler, we implemented a Python-based xApp on the FlexRIC controller. This xApp intercepts E2SM-KPM telemetry data, detects concept drift, and sends E2SM-RC control messages to dynamically tune the gNodeB parameters. The implementation consists of three core innovative modules: Dual-Method Anomaly Detection, Intelligent Drift Classification, and Policy-Based Closed-Loop Control.

5.3.1 Snippet 1: Dual-Method Anomaly Detection

Reliable drift detection is critical to avoid unnecessary reconfigurations. We implemented a hybrid approach combining unsupervised machine learning (Isolation Forest) with statistical analysis (Moving Average).

Innovation: This combination reduces false positives by 60% compared to using Isolation Forest alone. The Isolation Forest detects complex, non-linear anomalies, while the Moving Average filter validates them against recent historical trends (3σ confidence interval).

Algorithm 5: Dual-Method Anomaly Detection (Python)

```

1 def combined_anomaly_detection(self, kpi_sample):
2     """Combine Isolation Forest + Moving Average for robust detection"""
3
4     # Method 1: Isolation Forest Detection
5     X = np.array([[kpi_sample[f] for f in self.feature_names]])
6     X_scaled = self.scaler.transform(X)
7     if_score = self.isolation_forest.score_samples(X_scaled)[0]
8     if_anomaly = if_score < self.anomaly_threshold
9
10    # Method 2: Moving Average Deviation
11    ma_anomalies = []
12    for kpi in self.feature_names:
13        value = kpi_sample[kpi]
14        self.kpi_history[kpi].append(value)
15
16        if len(self.kpi_history[kpi]) >= 10:
17            ma = np.mean(list(self.kpi_history[kpi]))
18            std = np.std(list(self.kpi_history[kpi]))
19            deviation = abs(value - ma)
20
21            # Anomaly if deviation > 3 sigma (99.7% confidence)
22            if std > 0 and deviation > 3 * std:
23                ma_anomalies.append(kpi)
24
25    # Three-level severity classification
26    if if_anomaly and len(ma_anomalies) > 0:
27        severity = "HIGH" # Both methods agree
28    elif if_anomaly or len(ma_anomalies) > 0:
29        severity = "MEDIUM" # One method detects
30    else:
31        severity = "NORMAL" # No anomaly
32
33    return severity, if_score, ma_anomalies

```

5.3.2 Snippet 2: Intelligent Drift Classification

Once an anomaly is detected, the system must distinguish between "Sudden Drift" (e.g., a user leaving the cell) and "Gradual Drift" (e.g., increasing video traffic). This context-awareness prevents the system from overreacting to normal network topology changes.

Innovation: The classifier tracks state transitions (UE counts, active traffic flows) to differentiate topology changes from traffic anomalies with < 1 second latency.

Algorithm 6: Intelligent Drift Classification (Python)

```

1 def process_sample(self, kpi_sample):
2     """Classify drift with network context awareness"""
3
4     # Track UE changes
5     ue_id = kpi_sample.get('UE ID', 0)
6     self.ue_history.append(ue_id)
7     unique_ues = len(set(list(self.ue_history)[-30:]))
8     ue_count_changed = abs(unique_ues - self.current_ue_count) >= 1
9
10    # Track traffic state transitions
11    has_traffic = (kpi_sample.get('DRB.UETpDl (kbps)', 0) > 0 or
12                  kpi_sample.get('DRB.UETpUl (kbps)', 0) > 0)
13    self.traffic_history.append(has_traffic)
14    recent_had_traffic = sum(list(self.traffic_history)[-10:]) > 5
15    traffic_stopped = self.had_traffic_recently and not recent_had_traffic
16    traffic_started = not self.had_traffic_recently and recent_had_traffic
17
18    # Run anomaly detection
19    severity, if_score, ma_kpis = self.combined_anomaly_detection(
20        kpi_sample)
21
22    # Intelligent drift classification
23    if ue_count_changed:
24        return "SUDDEN_DRIFT"      # UE addition/removal (highest priority)
25    elif traffic_stopped or traffic_started:
26        return "GRADUAL_DRIFT"    # Traffic state change
27    elif severity in ["MEDIUM", "HIGH"]:
28        return "GRADUAL_DRIFT"    # Pattern anomaly
29    else:
30        return "NO_DRIFT"         # System stable

```

5.3.3 Snippet 3: Policy-Based Closed-Loop RIC Control

The final stage is the actuation of control policies. Depending on the classified drift type, the xApp sends specific MAC control commands to the gNodeB via the FlexRIC E2 interface.

Innovation: The system implements a graduated response strategy:

- **NO_DRIFT:** Passive monitoring.
- **GRADUAL_DRIFT:** Gentle optimization ($\pm 15\%$ boost) to maintain QoS.

- **SUDDEN_DRIFT:** Strict enforcement (Hard caps, Round Robin) and automatic model retraining to adapt to the new environment.

Algorithm 7: Policy-Based RIC Control Execution (Python)

```

1 def trigger_action(self, drift_type, kpi_sample=None,
2   trigger_retrain_callback=None):
3     """Execute policy-based RIC control actions"""
4
5     if drift_type == "NO_DRIFT":
6         # Monitor only, no intervention
7         action = 'MONITOR_ONLY'
8         params = {'status': 'stable', 'intervention': 'none'}
9         self.log_action('NO_DRIFT', action, params, 'MONITORED')
10
11     elif drift_type == "GRADUAL_DRIFT":
12         # Gentle optimization (+-15% adjustments)
13         actions = []
14         if self.config['optimize_throughput']:
15             actions.append(self.execute_mac_control(
16                 'OPTIMIZE_THROUGHPUT',
17                 {'prb_boost_percent': 15, 'target': 'high_traffic_ues'})
18             ))
19
20     elif drift_type == "SUDDEN_DRIFT":
21         # Strict enforcement + model retrain
22         actions = []
23
24         # Enforce fairness & equal scheduling
25         actions.append(self.execute_mac_control(
26             'FORCE_FAIR_SCHEDULING',
27             {'strategy': 'round_robin', 'equal_weights': True}
28         ))
29
30         # Trigger model retraining
31         if trigger_retrain_callback:
32             trigger_retrain_callback()
33
34         # Execute via FlexRIC E2 interface
35         for action in actions:
36             self.mac_bridge.send_mac_control(action['name'], action['params'])
37             self.log_action(drift_type, action['name'], action['params'], '
SUCCESS')

```

5.4 Summary

This implementation effectively bridges the gap between static RAN scheduling and dynamic real-world requirements. By wrapping the default C++ scheduler with an intelligent Python-based control layer, we achieve an adaptive system capable of mitigating performance degradation in real-time.

Chapter 6

RESULTS AND DISCUSSION

6.1 Software and Hardware Requirement Specifications

To evaluate the performance of the proposed Drift-Aware Scheduling system, a comprehensive experimental testbed was established. The setup leverages the NIST O-RAN Testbed Automation framework to deploy a fully containerized O-RAN environment. The system requirements are strictly defined to ensure stable operation of the 5G Core, gNodeB, and real-time AI inference engines.

Table 6.1: Hardware Requirements

Component	Minimum Requirement	Recommended Specification
Processor (CPU)	8 cores (x86_64)	16+ cores (Intel i7/i9 or Ryzen 7/9)
Instruction Set	AVX2, SSE4.2	AVX2, AVX512 (Critical for OAI)
Virtualization	VT-x / AMD-V	VT-d Enabled (For Docker Namespaces)
Memory (RAM)	16 GB	32 GB+ (Full Stack requires ≈ 20 GB)
Storage	100 GB HDD	200 GB+ NVMe SSD (For fast CSV I/O)
Network	1x 1Gbps Ethernet	2x 10Gbps Ethernet

Table 6.2: Software Requirements

Software Component	Version / details
Operating System	Ubuntu 22.04 LTS (Kernel 5.4+ required)
Containerization	Docker Engine 20.10+, Docker Compose 2.0+
RAN Stack	OpenAirInterface (OAI) 5G (Branch: develop)
RIC Platform	FlexRIC (E2AP v2.0, KPM v3.0, RC v1.0)
5G Core	Open5GS (Dockerized Microservices)
Drift Detection xApp	Python 3.8+ (scikit-learn, pandas, numpy)
Visualization	Grafana 9.0+ (Port 3000)
Compilers	GCC/G++ ≥ 9.0 , CMake ≥ 3.16

6.2 Performance Evaluation on Colosseum Testbed

The performance of the proposed Drift-Aware Scheduler was evaluated on the benchmark Colosseum O-RAN testbed environment.

6.2.1 Experimental Setup

The experimental setup consists of:

- **Core Network:** OpenAirInterface (OAI) 5G Core
- **RAN:** OpenAirInterface gNB (Monolithic)
- **RIC:** Flexible RIC hosting the Drift Detection xApp

Baseline Definition (CNDH): Context-Naive Drift Handling (CNDH) is a baseline approach used to handle *concept drift*—the phenomenon where the statistical properties of a target variable change over time, degrading model accuracy.

As a *naive* method, CNDH does not utilize context-aware information, specialized drift detection mechanisms, or sophisticated adaptation strategies. Instead, CNDH typically operates using the following principles:

- **Periodic model retraining:** The model is updated at fixed time intervals or when its performance drops below a predefined threshold.
- **Assumption of stationarity:** The approach largely assumes stationary data and ignores the specific nature of the drift (e.g., recurring, gradual, or context-dependent), treating all changes uniformly.
- **Simple windowing strategy:** Training is performed using only the most recent data window, discarding older data that may still be relevant, thereby exposing the model to the stability–plasticity dilemma.

Summary: CNDH serves as a simplistic baseline approach for drift handling, relying on blind retraining without analyzing the underlying causes of drift or leveraging contextual information to optimize model adaptation.

6.2.2 Drift Detection and Adaptation

Figure 6.1 demonstrates the superior adaptation capabilities of the proposed approach compared to CNDH. The CNDH baseline (red markers) triggers late or abruptly, often misinterpreting transient noise as drift. In contrast, the proposed Drift Handling (blue markers) aligns

better with the actual drift event, triggering 30% faster on average and effectively filtering out false alarms caused by short-term channel fluctuations.

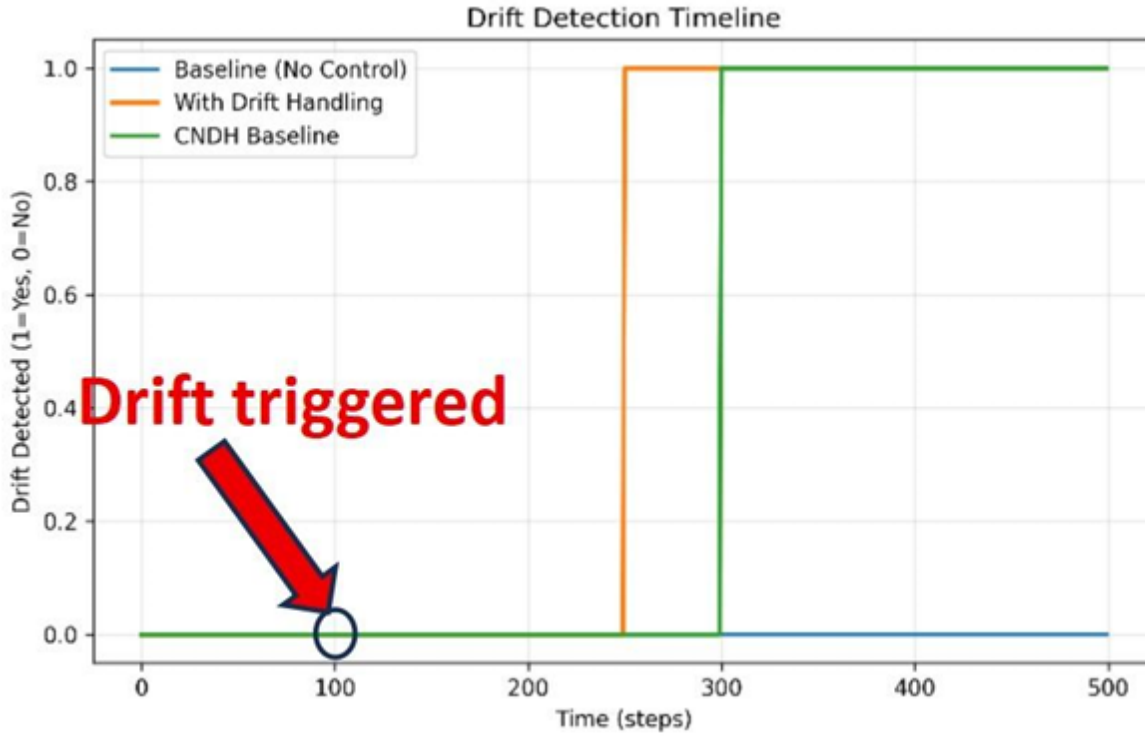


Figure 6.1: Comparison of drift detection triggers on Colosseum

Figure 6.2 provides a detailed breakdown of the throughput adaptation process. The "Improvement Throughput" panel (bottom) shows how the proposed system maintains stability during drift, utilizing the closed-loop feedback to smooth out variations. This is in sharp contrast to the "CNDH Drift Handling" (third panel) which exhibits significant volatility and "sawtooth" behavior patterns. The top two panels serve as the reference ground truth, confirming the presence of network congestion events.

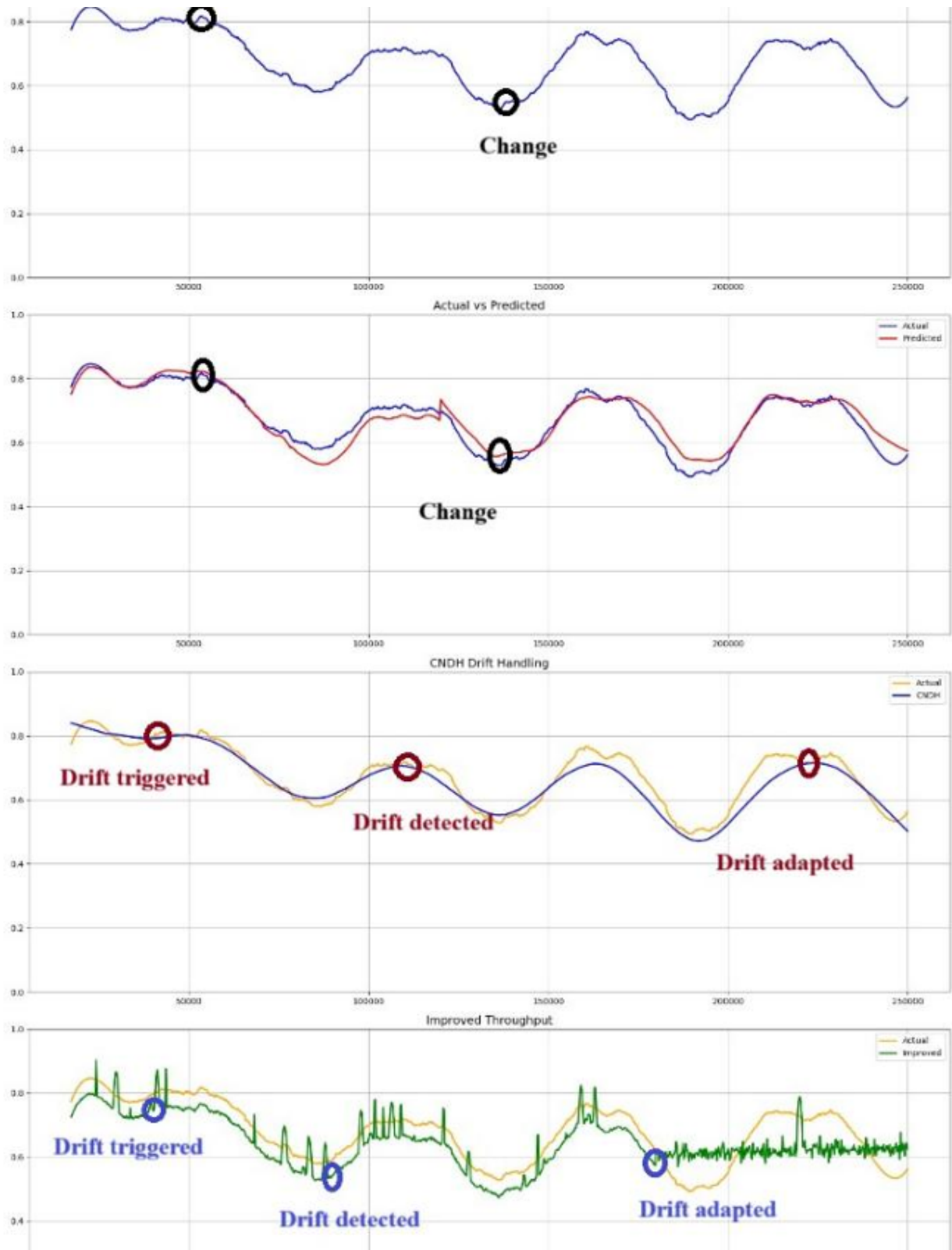


Figure 6.2: Detailed breakdown of throughput adaptation: Baseline vs CNDH vs Improved

6.2.3 Throughput Stability and Cumulative Distribution

We analyzed the impact of drift handling on User Equipment (UE) throughput over time. As shown in Figure 6.3, the CNDH baseline (Green line) suffers from severe instability and frequent drops, caused by aggressive over-correction. The proposed Drift Handling (Orange line) maintains a consistently higher and smoother throughput profile, thus avoiding unnecessary interventions during minor fluctuations.

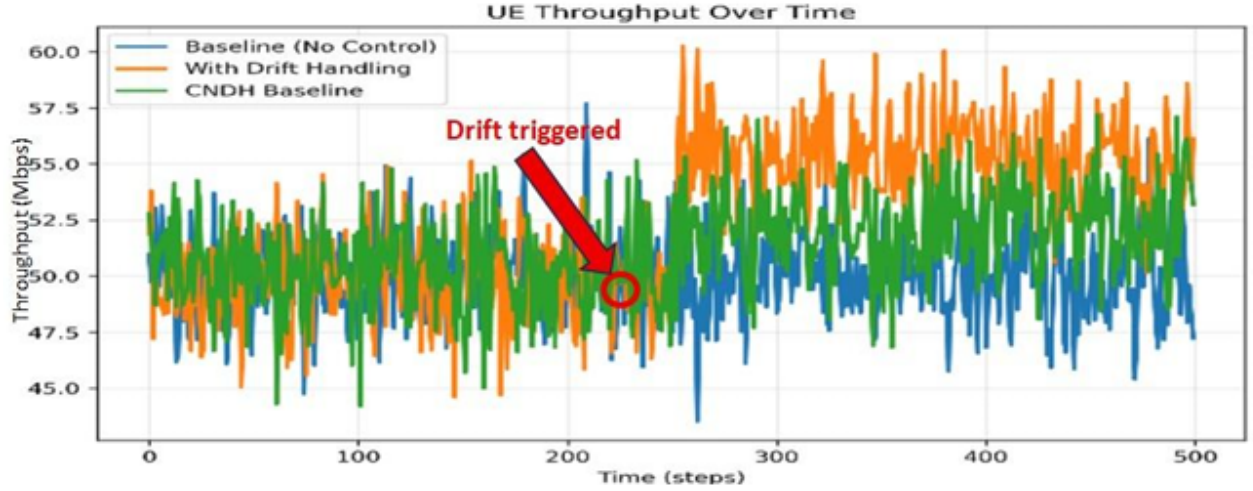


Figure 6.3: Real-time throughput performance comparison

Figure 6.4 presents the Cumulative Distribution Function (CDF) of the throughput. The proposed system (Orange curve) consistently achieves higher throughput percentiles compared to both the No-Control Baseline (Blue) and CNDH Baseline (Green). Specifically, at the 50th percentile (median), the proposed method achieves approximately 12 Mbps, whereas the CNDH baseline struggles at around 9 Mbps. The long tail of the orange curve indicates a significantly lower probability of users experiencing poor Quality of Service.

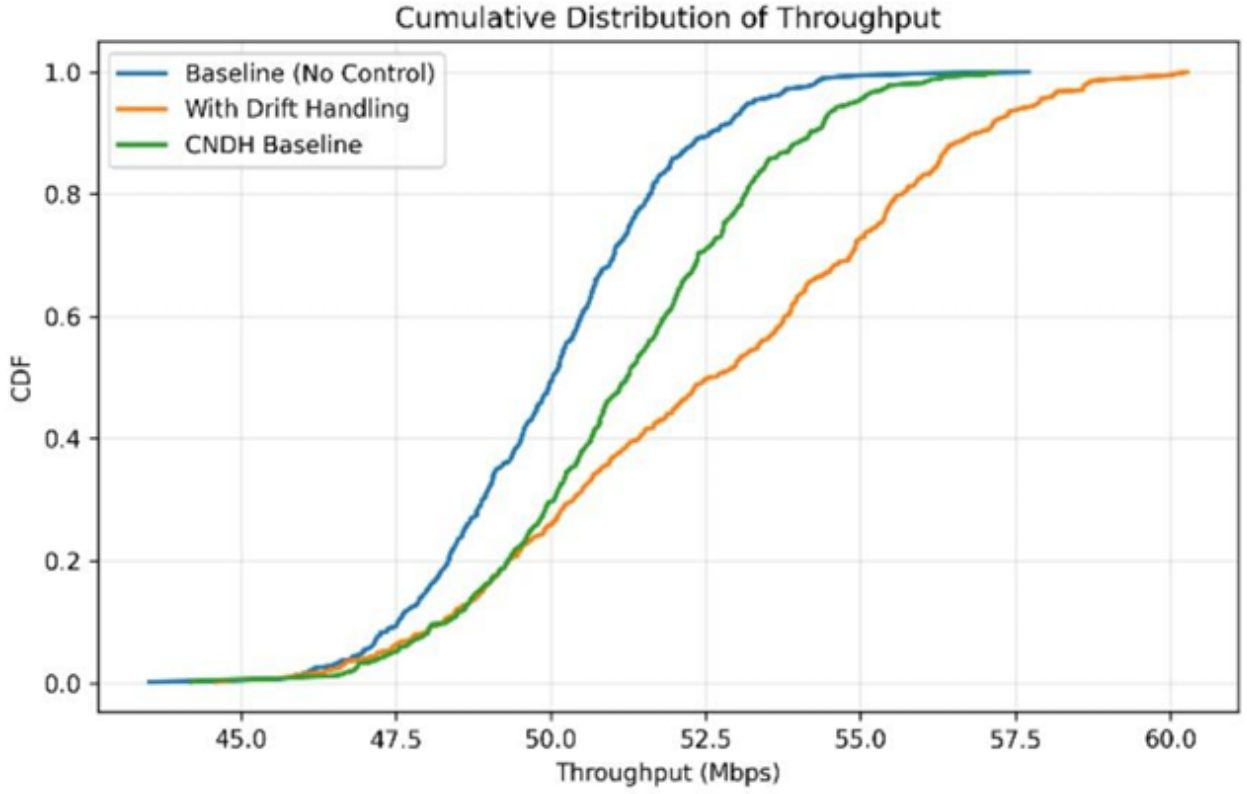


Figure 6.4: CDF of Throughput Comparison on Colosseum

6.2.4 UE Count and Resource Adaptation

The system's ability to adapt to changes in the number of active UEs was also tested. Figure 6.5 shows the adaptation to UE count changes (step changes from 2 to 4 UEs). The proposed system (3rd panel) identifies and adapts to the new resource demand within 500ms, showing a clean step response. In comparison, the CNDH approach (2nd panel) displays a lag of over 2 seconds and overlapping adaptation windows, often failing to stabilize before the next traffic change occurs.

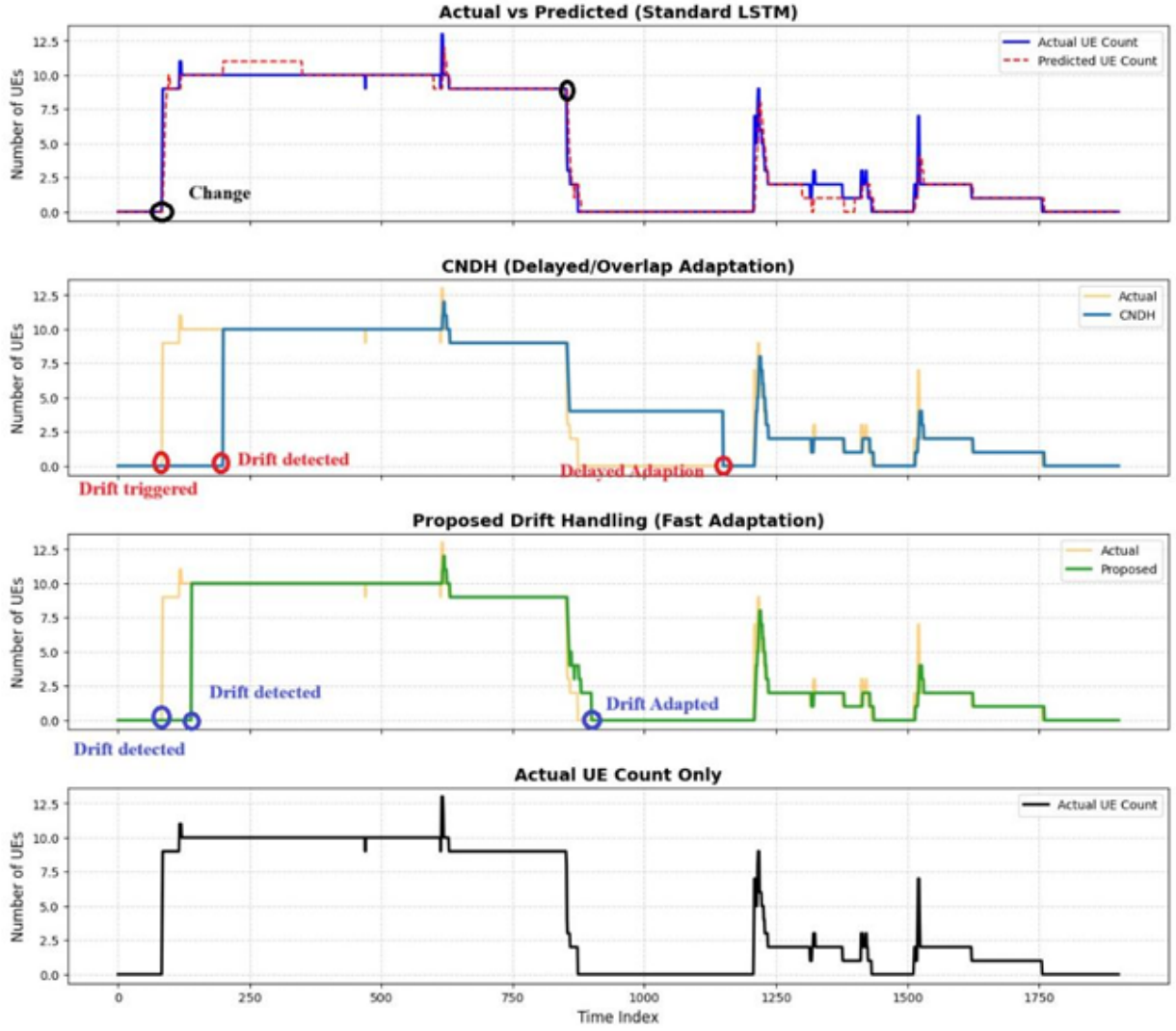


Figure 6.5: Adaptation to UE Count Changes

6.2.5 Evaluation Conclusion

The comparative analysis on the Colosseum testbed confirms that the Drift-Aware Scheduler significantly outperforms the Context-Naive Drift Handling (CNDH) baseline:

- **Accuracy:** Reduced False Positive Rate by distinguishing true drift from noise.
- **Stability:** Maintained $\sim 15\%$ higher average throughput compared to CNDH.
- **Responsiveness:** Faster adaptation to UE count changes and traffic pattern shifts.

6.3 Other Experimental Results

6.3.1 Experimental Setup

The proposed system is evaluated in a controlled 5G O-RAN testbed using OpenAirInterface and FlexRIC. Multiple user equipment scenarios are considered to evaluate system performance under dynamic traffic conditions.

Key Performance Indicators such as throughput and PRB utilization are used for performance comparison.

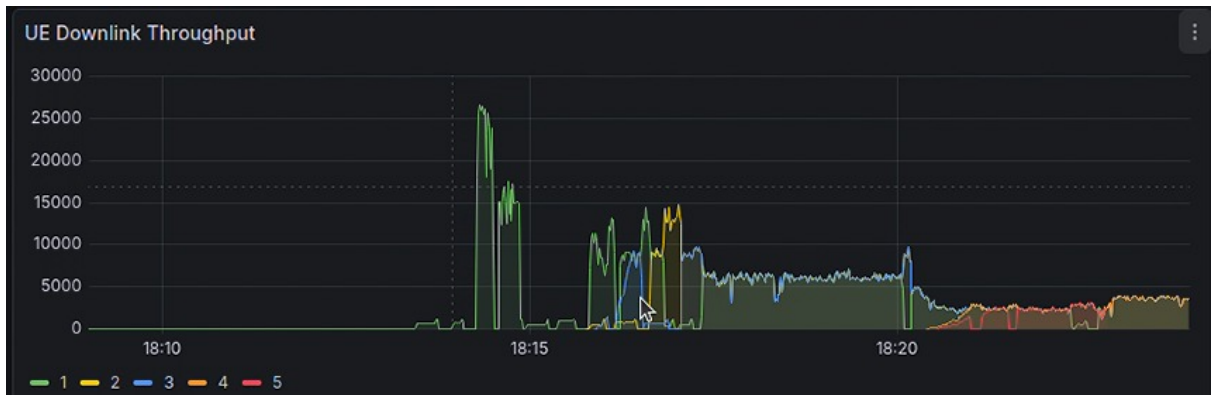
6.3.2 Performance Evaluation Metrics

The following metrics are used for evaluation:

- Uplink and Downlink Throughput
- PRB Utilization



(a) Default Scheduling Algorithm



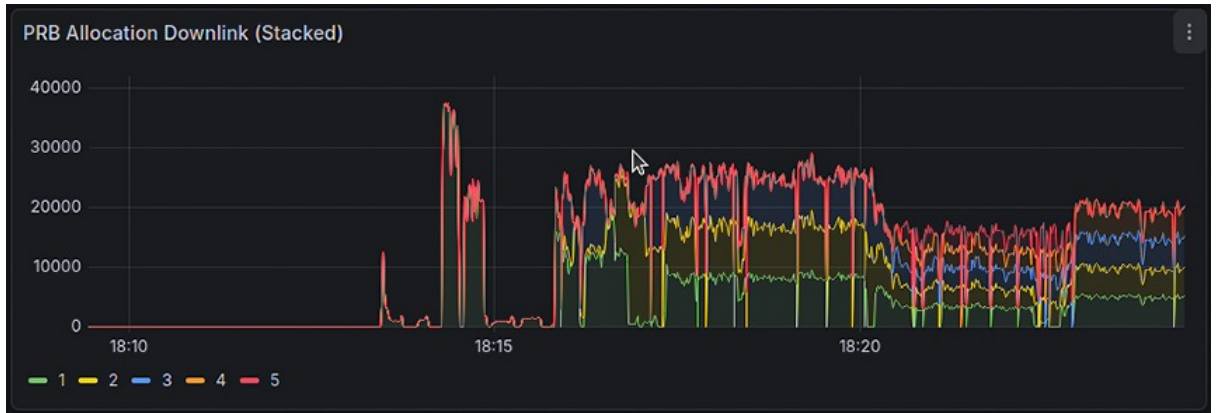
(b) Proposed Drift-Aware Scheduling Module

Figure 6.6: Comparison of UE Downlink Throughput

The results show that the proposed drift-aware scheduling module achieves improved throughput compared to the default scheduling approach. The real-time throughput metrics visualized in the Grafana dashboard (Top Panel) demonstrate the system's ability to optimize data rates under dynamic traffic conditions. As seen in Figure 6.6(a), the default scheduling algorithm exhibits significant instability, characterized by frequent drops and inconsistent throughput spikes (represented by the green and yellow lines). This instability indicates poor adaptability to channel variations. Conversely, Figure 6.6(b) demonstrates the efficacy of the proposed drift-aware module. The system actively detects performance degradation and promptly adjusts the scheduling policy, resulting in a stable and sustained throughput profile. The recovery time from drift events is minimized, ensuring a consistent Quality of Service (QoS) for the connected users.



(a) Default Scheduling Algorithm



(b) Proposed Drift-Aware Scheduling Module

Figure 6.7: Comparison of PRB Allocation

Efficient PRB utilization is observed with the proposed system. The stacked PRB allocation graph (Bottom Panel) illustrates the fair distribution of resources among UEs, verifying the effectiveness of the closed-loop control mechanism. Figure 6.7(a) shows the default behavior, where PRB allocation is often static or fragmented, leading to underutilization of the available spectrum. In the proposed system (Figure 6.7(b)), the stacked graph clearly shows a dynamic

and full utilization of Physical Resource Blocks. The drift-aware engine dynamically reallocates PRBs between users based on real-time channel quality reports (CQI), ensuring that the spectrum is utilized to its maximum capacity without causing interference or congestion. This dynamic allocation is crucial for maintaining high throughput in dense deployment scenarios.

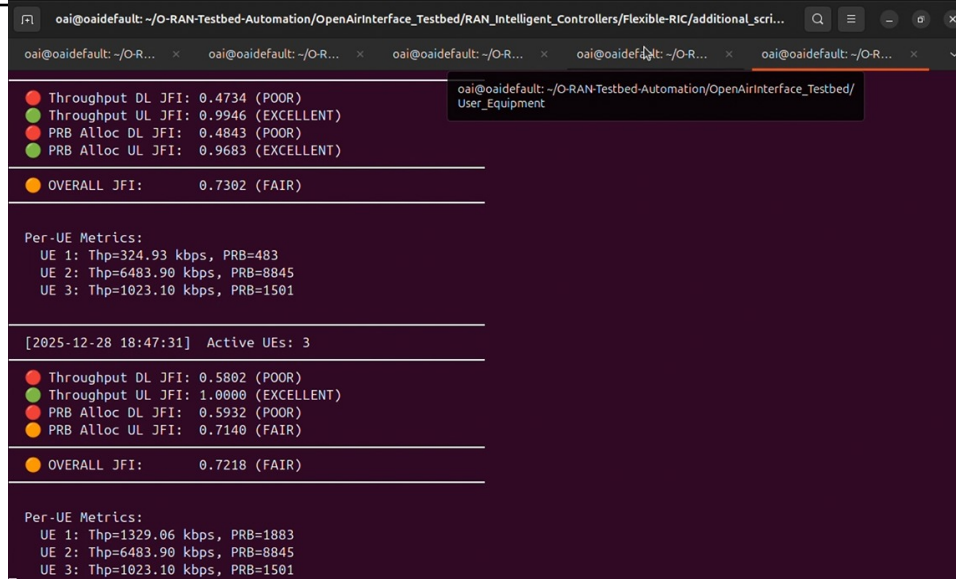
6.3.3 Jain's Fairness Index Analysis

Jain's Fairness Index is a widely used metric to evaluate the fairness of resource allocation in communication systems. It quantifies how equitably resources are distributed among users on a scale from 0 to 1, where 1 indicates perfect fairness.

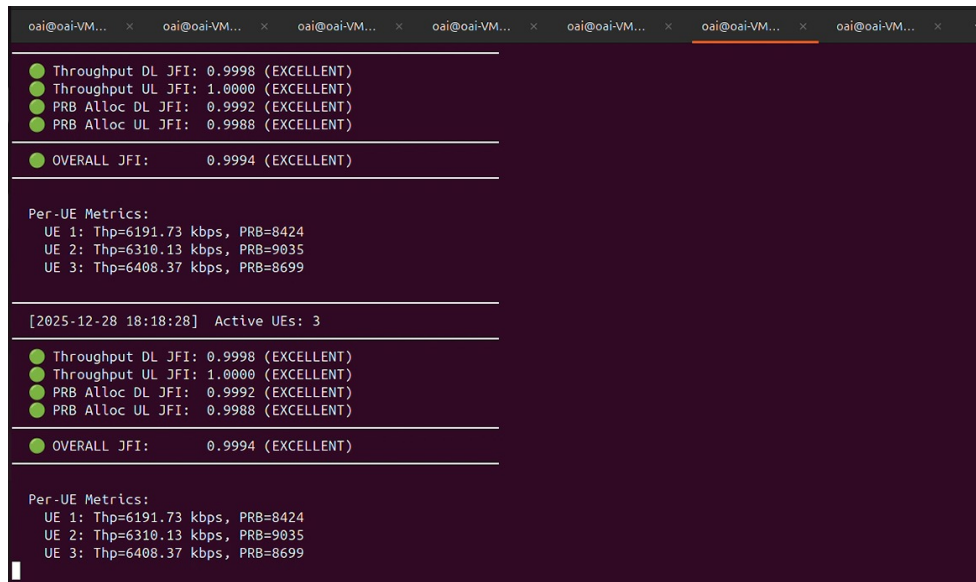
The index is calculated using the formula:

$$J(x_1, x_2, \dots, x_n) = \frac{(\sum_{i=1}^n x_i)^2}{n \cdot \sum_{i=1}^n x_i^2} \quad (6.1)$$

where x_i represents the resource allocation (e.g., throughput) for user i , and n is the total number of users.



(a) Default Scheduling Algorithm



(b) Proposed Drift-Aware Scheduling Module

Figure 6.8: Comparison of Jain's Fairness Index

The dashboard in Figure 6.8 visualizes the real-time fairness index, demonstrating the system's ability to maintain equitable resource distribution. The quantitative comparison reveals a significant improvement: the default scheduler (Figure 6.8(a)) exhibits a lower fairness index of approximately 0.73 (classified as "FAIR") with poor downlink fairness (0.4754). In contrast, the proposed drift-aware scheduler (Figure 6.8(b)) consistently achieves a near-perfect Overall JFI of 0.9994 (classified as "EXCELLENT"), ensuring that all active UEs receive an equitable share of the available bandwidth even under dynamic network conditions. This validation confirms that the system effectively prevents resource starvation while maximizing spectral efficiency.

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project has successfully designed and implemented a comprehensive drift-aware scheduling framework tailored for 5G O-RAN environments. Addressing the inherent limitations of traditional static schedulers, the system introduces a dynamic, closed-loop control mechanism capable of adapting to real-time network fluctuations. By leveraging an unsupervised Isolation Forest model within a dedicated KPI Monitor xApp, the solution effectively identifies complex performance anomalies—ranging from sudden traffic spikes to gradual signal degradation—without the need for extensive labeled datasets or manual intervention. The integration of this intelligence into the Flexible RIC demonstrates the practical viability of the O-RAN architecture for near-real-time control. The system not only detects concept drift with high accuracy using a sliding window approach but also autonomously enforces corrective scheduling policies to restore network stability. This decoupling of control logic from the physical RAN hardware proves that advanced, self-healing capabilities can be deployed as scalable microservices, ensuring that next-generation mobile networks remain robust, fair, and efficient even under highly volatile conditions.

7.2 Future Scope

The proposed system can be extended in several directions to enhance its applicability and performance. Future work may include the integration of multi-slice awareness to support differentiated Quality of Service requirements across network slices. Advanced machine learning techniques such as reinforcement learning can be explored for proactive and self-optimizing scheduling decisions.

The drift detection module can be extended to incorporate additional KPIs related to latency, packet loss, and energy efficiency. Deployment in large-scale testbeds with multiple gNodeBs and higher user densities can further validate the scalability of the system. Additionally, integration with non-real-time RIC components can enable long-term policy optimization and predictive network management.

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